



PRACTICE EXAMINATION

Mathematical Methods

2026

Answer booklet

- Worked solutions to Questions 1 to 11 (100 marks)
- Question booklet 1: Questions 1 to 6 (50 marks)
- Question booklet 2: Questions 7 to 11 (50 marks)
- Mark allocations are shown at the right of each awarded step
- *Marker's notes* highlight common errors and acceptable alternatives

Marks by topic

Topic	Questions	Marks
1 Further differentiation and applications	1, 3, 4, 8, 10, 11	40
2 Discrete random variables	5	8
3 Integral calculus	1, 3, 7	13
4 Logarithmic functions	1, 7, 8	11
5 Continuous random variables	2, 9	16
6 Sampling and confidence intervals	6, 9	12
Total		100

*Several questions draw on more than one topic; the marks above are allocated part by part to the topic that carries the main demand of that part.
Sampling distributions of the sample mean are Subtopic 5.3 and are counted under Topic 5.*

Written by TopMark Tutors as practice material for Stage 2 students. It is modelled on the format of the Stage 2 Mathematical Methods examination but is not produced, authorised, or endorsed by the SACE Board of South Australia. Answers were checked with a computer algebra system; probabilities are given to three significant figures unless an exact value is requested.

Question booklet 1 — solutions

Question 1

(8 marks) Differentiation rules; anti-differentiation

- (a)(i)
- $y = 3 \cos(4x) - \ln(x^3)$
- . Use the chain rule on the first term; note
- $\ln(x^3) = 3 \ln x$
- .

$$\frac{dy}{dx} = 3(-\sin(4x)) \times 4 - \frac{3}{x} = -12 \sin(4x) - \frac{3}{x} \quad [2 \text{ marks}]$$

One mark for $-12 \sin(4x)$ and one mark for $-\frac{3}{x}$.

- (ii)
- $y = x^3 \ln(2x)$
- . Product rule with
- $u = x^3$
- ,
- $v = \ln(2x)$
- , so
- $u' = 3x^2$
- and
- $v' = \frac{1}{2x} \times 2 = \frac{1}{x}$
- .

$$\frac{dy}{dx} = 3x^2 \ln(2x) + x^3 \times \frac{1}{x} = 3x^2 \ln(2x) + x^2 \quad [= x^2(3 \ln(2x) + 1)] \quad [2 \text{ marks}]$$

One mark for correct use of the product rule, one mark for the correct derivative.

- (iii)
- $y = \frac{5x^2 - 1}{e^{2x}}$
- . Quotient rule with
- $u = 5x^2 - 1$
- ,
- $v = e^{2x}$
- .

$$\frac{dy}{dx} = \frac{10xe^{2x} - (5x^2 - 1)(2e^{2x})}{(e^{2x})^2} = \frac{10x - 10x^2 + 2}{e^{2x}} \quad [= 2(-5x^2 + 5x + 1)e^{-2x}] \quad [2 \text{ marks}]$$

One mark for correct use of the quotient rule, one mark for the correct derivative. Writing $y = (5x^2 - 1)e^{-2x}$ and using the product rule is equally acceptable.

- (b) Reverse the chain rule for each term:

$$\int \left(e^{4x} + \frac{3}{2x+1} \right) dx = \frac{1}{4} e^{4x} + \frac{3}{2} \ln(2x+1) + c \quad [2 \text{ marks}]$$

One mark for $\frac{1}{4} e^{4x}$, one mark for $\frac{3}{2} \ln(2x+1)$.

Marker's note. The constant of integration is expected in part (b); its omission is penalised once across the whole paper. In (a)(ii) a very common slip is $\frac{d}{dx} \ln(2x) = \frac{1}{2x}$; the chain rule gives $\frac{1}{x}$. (Equivalently $\ln(2x) = \ln 2 + \ln x$.)

Question 2

(9 marks) Normal distribution; sampling distribution of the sample mean

$$T \sim N(42, 3.5^2).$$

- (a)(i)
- $\Pr(T < 40) = 0.2839 \dots = \mathbf{0.284}$
- [1 mark]

- (ii)
- $\Pr(38 \leq T \leq 45) = 0.6777 \dots = \mathbf{0.678}$
- [1 mark]

- (b) Require
- $\Pr(T \leq k) = 0.95$
- , so
- k
- is the inverse-normal value:

$$k = 42 + 1.6449 \times 3.5 = 47.757 \dots \quad [1 \text{ mark: correct inverse normal set-up}]$$

$$k = \mathbf{47.8} \text{ minutes} \quad [1 \text{ mark: answer}]$$

- (c)(i) For a sample of size
- $n = 6$
- :

$$\mu_{\bar{T}_6} = \mu = \mathbf{42} \text{ minutes} \quad [1 \text{ mark}]$$

$$\sigma_{\bar{T}_6} = \frac{\sigma}{\sqrt{n}} = \frac{3.5}{\sqrt{6}} = 1.4288 \dots = \mathbf{1.43} \text{ minutes} \quad [1 \text{ mark}]$$

- (ii) $\Pr(\bar{T}_6 > 44) = 0.0808$ (using $N(42, 1.4289^2)$) [1 mark]
- (d) Two marks, for example:
- $\bar{T}_6$ is the mean of 6 observations, so its sampling distribution has the same mean (42) but a *smaller* standard deviation, $\frac{3.5}{\sqrt{6}} \approx 1.43$, compared with 3.5 for a single trip. [1 mark]
 - Sample means are therefore clustered more tightly about 42, so a value above 44 is far less likely for $\bar{T}_6$ than for T : $\Pr(\bar{T}_6 > 44) = 0.0808$ whereas $\Pr(T > 44) = 0.284$. [1 mark]

Marker's note. In (c)(ii), a student who carries the rounded value $\sigma_{\bar{T}_6} = 1.43$ into the calculation obtains 0.0810; both 0.0808 and 0.0810 receive the mark. Part (d) asks for a reason, not a calculation — a response that only quotes the two probabilities without linking the difference to the reduced standard deviation of the sampling distribution earns at most one mark.

Question 3**(9 marks)** *Exponential functions; rectilinear motion; integration*

$$v_D(t) = 12 - 12e^{-0.5t}.$$

- (a) Acceleration is the derivative of velocity:

$$a(t) = v'_D(t) = -12 \times (-0.5)e^{-0.5t} = 6e^{-0.5t} \quad [1 \text{ mark}]$$

$$a(2) = 6e^{-1} = 2.2072 \dots = \mathbf{2.21} \text{ m s}^{-2} \quad [1 \text{ mark}]$$

- (b) For every t , $e^{-0.5t} > 0$, so $12e^{-0.5t} > 0$ and hence $v_D(t) = 12 - 12e^{-0.5t} < 12$. [1 mark]

- (c) Displacement is the anti-derivative of velocity:

$$h(t) = \int (12 - 12e^{-0.5t}) dt = 12t - \frac{12}{-0.5}e^{-0.5t} + c = 12t + 24e^{-0.5t} + c \quad [2 \text{ marks}]$$

$$h(0) = 0: \quad 0 + 24e^0 + c = 0 \Rightarrow c = -24 \quad [1 \text{ mark}]$$

$$\therefore h(t) = 12t + 24e^{-0.5t} - 24 \quad \text{as required.}$$

One mark for the term $12t$, one mark for the term $+24e^{-0.5t}$, one mark for using the initial condition to find c .

- (d) $h(10) = 120 + 24e^{-5} - 24 = 96.1617 \dots = \mathbf{96.2}$ metres [1 mark]

- (e) Solve $12t + 24e^{-0.5t} - 24 = 50$ using electronic technology:

$$12t + 24e^{-0.5t} = 74 \quad [1 \text{ mark: correct equation}]$$

$$t = 6.0705 \dots = \mathbf{6.07} \text{ seconds} \quad [1 \text{ mark: answer}]$$

Marker's note. In (c) the two most common errors are integrating $-12e^{-0.5t}$ to $-6e^{-0.5t}$ (dividing by -0.5 rather than multiplying by $\frac{1}{-0.5}$) and omitting the constant of integration, which then makes the initial condition impossible to apply. A “show that” question requires the constant to be evaluated explicitly.

Question 4**(7 marks)** *Differentiation from first principles; tangents*

(a) $f(x) = \frac{1}{2x+1}.$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{1}{2(x+h)+1} - \frac{1}{2x+1} \right] \quad [1 \text{ mark: definition}]$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{(2x+1) - (2x+2h+1)}{(2x+2h+1)(2x+1)} && [1 \text{ mark: common denominator}] \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{-2h}{(2x+2h+1)(2x+1)} = \lim_{h \rightarrow 0} \frac{-2}{(2x+2h+1)(2x+1)} && [1 \text{ mark: simplify and cancel } h] \\
&= \frac{-2}{(2x+1)^2} && [1 \text{ mark: evaluate the limit}]
\end{aligned}$$

(b) At $x = 1$: $f(1) = \frac{1}{3}$ and $f'(1) = \frac{-2}{3^2} = -\frac{2}{9}$. [1 mark]

$$y - \frac{1}{3} = -\frac{2}{9}(x - 1) \Rightarrow y = -\frac{2}{9}x + \frac{5}{9} \quad (\text{or } 2x + 9y - 5 = 0) \quad [1 \text{ mark}]$$

(c) $f'(x) = \frac{-2}{(2x+1)^2}$. Since $(2x+1)^2 > 0$ for all $x \neq -\frac{1}{2}$, $f'(x) < 0$ everywhere in the domain and is never equal to zero. A stationary point requires $f'(x) = 0$, so there are none. [1 mark]

Marker's note. Marks in (a) are for the *process*, so the limit notation must be carried through every line until the limit is taken — writing the lines without $\lim_{h \rightarrow 0}$ and only inserting it at the end loses a mark. Quoting the quotient-rule answer in (a) earns nothing, because the question specifies first principles.

Question 5

(8 marks) *Discrete random variables; the binomial distribution*

(a) The probabilities of a probability distribution sum to 1:

$$0.45 + 0.30 + a + 0.05 = 1 \Rightarrow a = 1 - 0.80 = 0.20 \quad \text{as required.} \quad [1 \text{ mark}]$$

(b) $E(X) = \sum x p(x)$:

$$E(X) = 0(0.45) + 1(0.30) + 2(0.20) + 3(0.05) \quad [1 \text{ mark: correct set-up}]$$

$$= 0.30 + 0.40 + 0.15 = \mathbf{0.85} \quad [1 \text{ mark: answer}]$$

(c) $\sigma_X = \sqrt{\sum [x - \mu_X]^2 p(x)}$, or equivalently $\sqrt{E(X^2) - \mu_X^2}$:

$$E(X^2) = 0 + 1(0.30) + 4(0.20) + 9(0.05) = 1.55 \quad [1 \text{ mark: correct method}]$$

$$\text{Var}(X) = 1.55 - 0.85^2 = 0.8275 \Rightarrow \sigma_X = \sqrt{0.8275} = 0.90967 \dots = \mathbf{0.910} \quad [1 \text{ mark: answer}]$$

(d)(i) “At least one token” means $X \geq 1$, with probability $1 - 0.45 = 0.55$, over $n = 20$ independent purchases:

$$Y \sim B(20, 0.55) \quad [1 \text{ mark}]$$

(ii) $\Pr(Y \geq 12) = 1 - \Pr(Y \leq 11) = 0.41430 \dots = \mathbf{0.414}$ [1 mark]

(iii) $E(Y) = np = 20 \times 0.55 = \mathbf{11}$ [1 mark]

Marker's note. In (d)(i) the most frequent error is using $p = 0.45$ (the probability of *no* token). In (d)(ii), $\Pr(Y \geq 12)$ must exclude $Y = 11$: subtracting $\Pr(Y \leq 12)$ instead gives 0.252 and earns no mark.

Question 6**(9 marks)** *Population proportions; confidence intervals*

(a) $\hat{p} = \frac{145}{250} = \mathbf{0.58}$ [1 mark]

(b) $\hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ with $z = 1.96$:

$$0.58 \pm 1.96 \sqrt{\frac{0.58 \times 0.42}{250}} = 0.58 \pm 1.96 (0.031215) = 0.58 \pm 0.06118$$
 [1 mark: correct substitution]

$$\mathbf{0.519 \leq p \leq 0.641}$$
 [1 mark: correct limits]

(c)(i) **Yes** [1 mark]

(ii) The whole confidence interval lies above 0.5 (the lower limit is $0.519 > 0.5$), so with 95% confidence the population proportion of supporters exceeds one half. [1 mark]

(d) The interval would be about **half as wide**. [1 mark]

The margin of error is $z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, which is inversely proportional to $\sqrt{n}$. Multiplying n by 4 multiplies $\sqrt{n}$ by 2, so the margin of error — and hence the width — is divided by 2. [1 mark: justification]

(e) A width of less than 0.05 means a margin of error of less than 0.025:

$$1.96 \sqrt{\frac{0.58 \times 0.42}{n}} < 0.025$$
 [1 mark: correct inequality]

$$n > \frac{1.96^2 \times 0.58 \times 0.42}{0.025^2} = 1497.3 \dots \Rightarrow n = \mathbf{1498}$$
 [1 mark: answer]

Marker's note. In (e) the answer must be rounded *up* to the next whole number: $n = 1497$ gives a width of 0.050005, which is not less than 0.05. Answers of 1497 or 1500 do not receive the second mark.

Question booklet 2 — solutions

Question 7

(9 marks) Area under a curve; logarithmic integration

$$f(x) = \frac{8}{x+1}, x \geq 0.$$

- (a) Four rectangles of width 1 on $[0, 4]$, each with its height taken at the *right-hand* end of the sub-interval (because f is decreasing): heights $f(1) = 4$, $f(2) = \frac{8}{3}$, $f(3) = 2$, $f(4) = \frac{8}{5}$. Every rectangle sits under the curve. [1 mark]

- (b) Each rectangle has width $\frac{4}{4} = 1$:

$$L_4 = 1 \times [f(1) + f(2) + f(3) + f(4)] = 4 + \frac{8}{3} + 2 + \frac{8}{5}$$

[1 mark: correct heights and width]

$$= 10.2666\dots = \mathbf{10.3} \text{ square units}$$

[1 mark: answer]

- (c) $f'(x) = \frac{-8}{(x+1)^2} < 0$, so f is decreasing on $[0, 4]$. [1 mark]

Taking the height at the right-hand end of each sub-interval therefore uses the *smallest* value of f on that sub-interval, so every rectangle lies entirely below the curve and the sum of their areas must be less than the area of the shaded region. [1 mark]

- (d) Since $\frac{d}{dx} \ln(x+1) = \frac{1}{x+1}$:

$$\int_0^4 \frac{8}{x+1} dx = \left[8 \ln(x+1) \right]_0^4$$

[1 mark: correct anti-derivative]

$$= 8 \ln 5 - 8 \ln 1 = \mathbf{8 \ln 5} \text{ square units } (\approx 12.9)$$

[1 mark: exact value]

- (e) Use the same anti-derivative with an unknown upper limit:

$$\int_0^b \frac{8}{x+1} dx = 8 \ln(b+1) = 20$$

[1 mark: correct equation]

$$\ln(b+1) = \frac{5}{2} \Rightarrow b+1 = e^{5/2} \Rightarrow \mathbf{b = e^{5/2} - 1} (\approx 11.2)$$

[1 mark: exact answer]

Marker's note. Parts (d) and (e) ask for *exact* values, so 12.9 and 11.2 alone do not earn the final mark in each part. Note also that, since $x \geq 0$ throughout the region, $x+1 > 0$ and the modulus signs in $\int \frac{8}{x+1} dx = 8 \ln|x+1| + c$ are unnecessary.

Question 8

(10 marks) Calculus of logarithmic functions; the second derivative

$$f(x) = x^2 \ln x, x > 0.$$

- (a) Product rule with $u = x^2$, $v = \ln x$:

$$f'(x) = 2x \ln x + x^2 \times \frac{1}{x}$$

[1 mark: product rule]

$$= 2x \ln x + x = x(2 \ln x + 1) \quad \text{as required.}$$

[1 mark]

- (b) Stationary points occur where $f'(x) = 0$:

$$x(2 \ln x + 1) = 0. \text{ Since } x > 0, x \neq 0, \text{ so } 2 \ln x + 1 = 0. \quad [1 \text{ mark}]$$

$$\ln x = -\frac{1}{2} \Rightarrow x = e^{-1/2} \quad [1 \text{ mark}]$$

$$y = (e^{-1/2})^2 \ln(e^{-1/2}) = e^{-1} \times \left(-\frac{1}{2}\right) = -\frac{1}{2e} \quad [1 \text{ mark}]$$

$$\text{Stationary point: } \left(e^{-1/2}, -\frac{1}{2e}\right) \approx (0.607, -0.184)$$

- (c) Differentiate $f'(x) = 2x \ln x + x$ using the product rule on $2x \ln x$:

$$f''(x) = 2 \left(\ln x + x \times \frac{1}{x} \right) + 1 \quad [1 \text{ mark}]$$

$$= 2 \ln x + 2 + 1 = 2 \ln x + 3 \quad \text{as required.} \quad [1 \text{ mark}]$$

- (d)(i) At $x = e^{-1/2}$: $f''(e^{-1/2}) = 2 \left(-\frac{1}{2}\right) + 3 = 2 > 0$, so the curve is concave up there and the stationary point is a **local minimum**. [1 mark]

- (ii) A point of inflection requires $f''(x) = 0$:

$$2 \ln x + 3 = 0 \Rightarrow \ln x = -\frac{3}{2} \quad [1 \text{ mark}]$$

$$x = e^{-3/2} \approx 0.223 \quad [1 \text{ mark}]$$

f'' is increasing, so it changes from negative to positive at $x = e^{-3/2}$: the concavity does change and this is a genuine point of inflection.

Marker's note. In (b) the step “ $x > 0$, therefore reject $x = 0$ ” is worth stating explicitly — the domain restriction is what makes $x = e^{-1/2}$ the only stationary point. Exact coordinates are required; decimal answers alone lose the final mark of (b).

Question 9

(10 marks) *Continuous random variables; confidence interval for a mean*

$$f(x) = kx(8 - x) \text{ for } 0 \leq x \leq 8.$$

- (a) The total area under a probability density function is 1:

$$\int_0^8 kx(8 - x) dx = 1 \Rightarrow k \int_0^8 (8x - x^2) dx = 1 \quad [1 \text{ mark: correct condition}]$$

$$k \left[4x^2 - \frac{x^3}{3} \right]_0^8 = k \left(256 - \frac{512}{3} \right) = k \times \frac{256}{3} \quad [1 \text{ mark: correct integration}]$$

$$\frac{256k}{3} = 1 \Rightarrow k = \frac{3}{256} \quad \text{as required.} \quad [1 \text{ mark}]$$

- (b) Integrate the density function from 0 to 3:

$$\Pr(X \leq 3) = \frac{3}{256} \int_0^3 (8x - x^2) dx = \frac{3}{256} (36 - 9) = \frac{81}{256} = 0.31640\dots = \mathbf{0.316} \quad [1 \text{ mark}]$$

- (c) $\mu_X = \int_{-\infty}^{\infty} xf(x) dx$:

$$\mu_X = \frac{3}{256} \int_0^8 x(8x - x^2) dx = \frac{3}{256} \int_0^8 (8x^2 - x^3) dx \quad [1 \text{ mark: correct integral}]$$

$$= \frac{3}{256} \left[\frac{8x^3}{3} - \frac{x^4}{4} \right]_0^8 = \frac{3}{256} \left(\frac{4096}{3} - 1024 \right) \quad [1 \text{ mark: correct integration}]$$

$$= \frac{3}{256} \times \frac{1024}{3} = \frac{1024}{256} = 4 \quad \text{as required.} \quad [1 \text{ mark}]$$

(d)(i) $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$ with $\bar{x} = 4.42$, $\sigma = 1.789$, $n = 40$, $z = 1.96$:

$$4.42 \pm 1.96 \times \frac{1.789}{\sqrt{40}} = 4.42 \pm 1.96(0.28287) = 4.42 \pm 0.5544 \quad [1 \text{ mark: correct substitution}]$$

$$3.87 \leq \mu \leq 4.97 \text{ minutes} \quad [1 \text{ mark: correct limits}]$$

(ii) **No.** The value $\mu = 4$ lies *inside* the confidence interval $[3.87, 4.97]$, so a population mean of 4 minutes remains consistent with the sample. The interval therefore does not support the claim of an increase with 95% confidence. [1 mark]

Marker's note. The “show that” parts (a) and (c) must be done algebraically — a numerical integration on a calculator that simply returns 1 or 4 earns no marks. In (d)(ii), a correct “No” with no reference to 4 lying inside the interval does not earn the mark.

Question 10

(10 marks) *Trigonometric functions and their derivatives; modelling*

$$D(t) = 3.2 + 1.4 \sin\left(\frac{\pi t}{6}\right), 0 \leq t \leq 24.$$

(a) Chain rule: $D'(t) = 1.4 \times \frac{\pi}{6} \cos\left(\frac{\pi t}{6}\right) = \frac{7\pi}{30} \cos\left(\frac{\pi t}{6}\right) \quad (\approx 0.733 \cos(\frac{\pi t}{6}))$ [1 mark]

(b) Substitute $t = 4$, noting that $\cos(\frac{2\pi}{3}) = -\frac{1}{2}$:

$$D'(4) = \frac{7\pi}{30} \cos\left(\frac{2\pi}{3}\right) = \frac{7\pi}{30} \times \left(-\frac{1}{2}\right) = -\frac{7\pi}{60} = -0.367 \quad [1 \text{ mark}]$$

Interpretation: at $t = 4$ (4 a.m.) the depth of the water at the jetty is *decreasing* at a rate of 0.367 metres per hour. [1 mark]

(c) Maxima and minima occur where $D'(t) = 0$:

$$\cos\left(\frac{\pi t}{6}\right) = 0 \quad [1 \text{ mark}]$$

$$\frac{\pi t}{6} = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \Rightarrow t = 3 \text{ or } t = 9 \quad [1 \text{ mark}]$$

$$D(3) = 3.2 + 1.4 = 4.6 \text{ m — maximum at } t = 3; \quad D(9) = 3.2 - 1.4 = 1.8 \text{ m — minimum at } t = 9. \quad [1 \text{ mark}]$$

(d) The boat can leave when $D(t) \geq 4$:

$$3.2 + 1.4 \sin\left(\frac{\pi t}{6}\right) \geq 4 \Rightarrow \sin\left(\frac{\pi t}{6}\right) \geq \frac{4}{7} \quad [1 \text{ mark: correct inequality}]$$

$$\frac{\pi t}{6} = \sin^{-1}\left(\frac{4}{7}\right) = 0.60825 \Rightarrow t = 1.1617$$

$$\frac{\pi t}{6} = \pi - 0.60825 = 2.53334 \Rightarrow t = 4.8383 \quad [1 \text{ mark: both end points}]$$

So $D \geq 4$ for $1.1617 \leq t \leq 4.8383$, a window of length 3.6767 hours. The period of D is $\frac{2\pi}{\pi/6} = 12$ hours, so this window occurs *twice* in $0 \leq t \leq 24$ (again for $13.1617 \leq t \leq 16.8383$).

Total time = $2 \times 3.6767 = 7.3533 \dots = 7.35$ hours (about 7 hours 21 minutes).

[1 mark: doubling for the second cycle, and the final answer]

(e) New amplitude = $\frac{5.0 - 1.4}{2} = 1.8$; new mean depth = $\frac{5.0 + 1.4}{2} = 3.2$ (unchanged).

$$D(t) = 3.2 + 1.8 \sin\left(\frac{\pi t}{6}\right) \quad [1 \text{ mark}]$$

Marker's note. The single most common error in (d) is giving 3.68 hours — the length of *one* high-water window — and forgetting that the 24-hour period contains two complete 12-hour cycles. A second common error is using $\sin^{-1}\left(\frac{4}{7}\right)$ in degrees; the model is in radians.

Question 11**(11 marks) Conjecture and proof; exponential functions; optimisation**

$f_n(x) = x^n e^{-x}$, $x \geq 0$, n a positive integer.

(a) Product rule with $u = x^2$, $v = e^{-x}$:

$$f_2'(x) = 2xe^{-x} + x^2(-e^{-x}) = 2xe^{-x} - x^2e^{-x} \quad [1 \text{ mark: product rule}]$$

$$= x(2 - x)e^{-x} \quad \text{as required.} \quad [1 \text{ mark: factorisation}]$$

(b) **Conjecture:** for $x > 0$, the maximum turning point of $y = f_n(x)$ occurs at $x = n$. [1 mark]

(c) **Proof.** Differentiate $f_n(x) = x^n e^{-x}$ using the product rule:

$$f_n'(x) = nx^{n-1}e^{-x} + x^n(-e^{-x}) = nx^{n-1}e^{-x} - x^n e^{-x} \quad [1 \text{ mark: product rule}]$$

$$= x^{n-1}e^{-x}(n - x) \quad [1 \text{ mark: factorisation}]$$

For $x > 0$ we have $x^{n-1} > 0$ and $e^{-x} > 0$, so $f_n'(x) = 0 \iff n - x = 0 \iff x = n$; this is the only stationary point for $x > 0$. [1 mark]

Sign of f_n' : for $0 < x < n$, $(n - x) > 0$ so $f_n'(x) > 0$ and f_n is increasing; for $x > n$, $(n - x) < 0$ so $f_n'(x) < 0$ and f_n is decreasing. The derivative changes from positive to negative at $x = n$, so $x = n$ is a maximum turning point. [1 mark]

Hence the conjecture is proved for every positive integer n . ■

(d)(i) Substituting $x = n$:

$$f_n(n) = n^n e^{-n} = \frac{n^n}{e^n} = \left(\frac{n}{e}\right)^n \quad \text{as required.} \quad [1 \text{ mark}]$$

(ii) Require $\left(\frac{n}{e}\right)^n > 100$. Evaluating for successive integers:

n	3	4	5	6
$\left(\frac{n}{e}\right)^n$	1.34	4.69	21.1	116

[1 mark: correct inequality set up]

[1 mark: evidence (table or graph) showing $n = 5$ fails and $n = 6$ succeeds]

For $n \geq 3$, $\frac{n}{e} > 1$ and both the base and the exponent increase with n , so $\left(\frac{n}{e}\right)^n$ is increasing. Since $\left(\frac{5}{e}\right)^5 = 21.1 < 100$ and $\left(\frac{6}{e}\right)^6 = 116 > 100$, the smallest such integer is $n = 6$. [1 mark: answer]

Marker's note. Part (c) is assessed on the *reasoning*, not just the algebra. Two marks are commonly lost: stating $f_n'(x) = 0$ at $x = n$ without explaining why $x^{n-1}e^{-x} \neq 0$ for $x > 0$, and omitting the sign test (or a second-derivative test) that establishes the stationary point is a *maximum*. A verification for a few values of n is not a proof.

Mark distribution

Booklet 1	Q1	Q2	Q3	Q4	Q5	Q6	Total
Marks	8	9	9	7	8	9	50

Booklet 2	Q7	Q8	Q9	Q10	Q11	Total
Marks	9	10	10	10	11	50

Paper total: 100 marks · 130 minutes